Retail Demand Forecasting & Forecast Error Analysis

A recruiter-facing case study demonstrating demand analysis, practical forecasting, error diagnosis, and business storytelling.

BUSINESS OBJECTIVE

Support retail demand planning by forecasting short-term demand, evaluating forecast performance, and identifying Product × Store segments where demand is more difficult to predict.

01 · WHAT HAPPENED?	*02 · HOW ACCURATE?*	*03 · WHERE / WHY?*
What does retail demand look like and what patterns exist?	How accurately can short-term demand be forecast?	Where are errors highest, and what demand patterns are associated with them?

Analysis flow

Data Preparation → Demand Analysis → Short-Term Forecasting → Forecast Evaluation → Error Diagnosis → Planning Implications

Tools: Python · Pandas · NumPy · Matplotlib · Seaborn **Dataset:** Walmart M5 retail demand data **Forecast grain:** Product × Store

1. From Raw Data to Analysis

The project uses four notebooks to move from data quality and structure checks to business analysis and forecast diagnosis.

DATA PREPARATION	*TRANSFORMATION*	*VALIDATION*
Validated dataset grains, identifiers, relationships, dates, demand ranges, prices and referential integrity. Converted calendar dates to datetime.	Converted wide daily-sales data to Product × Store × Day format. Historical analysis: 58,327,370 rows. Evaluation period: 853,720 rows.	Transformation preserved demand totals, product-store coverage and daily structure. No negative demand or duplicate product-store-day records were identified.

Forecasting scenario

Historical data	Forecast cutoff	Evaluation	Grain
Jan 29, 2011 – Apr 24, 2016	Apr 24, 2016	Apr 25 – May 22, 2016 28 days	30,490 Product × Store series

Why this matters

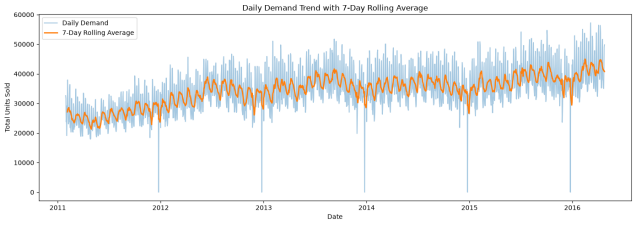
• The preparation stage establishes a reliable analytical grain before forecasting.

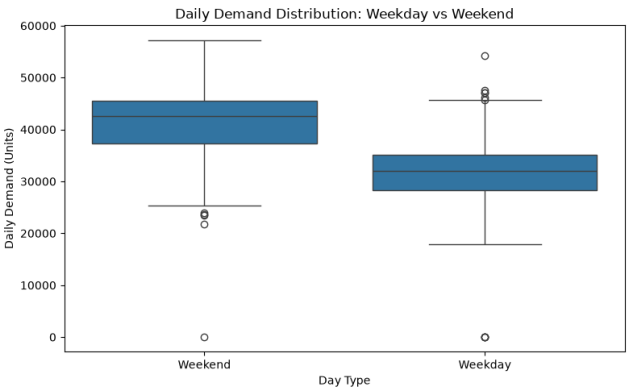
• The forecasting setup avoids using future actual demand when generating forecasts.

• The analysis retains unusual demand and price observations for diagnosis rather than automatically treating them as data errors.

2. WHAT HAPPENED? — Demand Patterns

Historical demand analysis establishes the business context before forecasting.



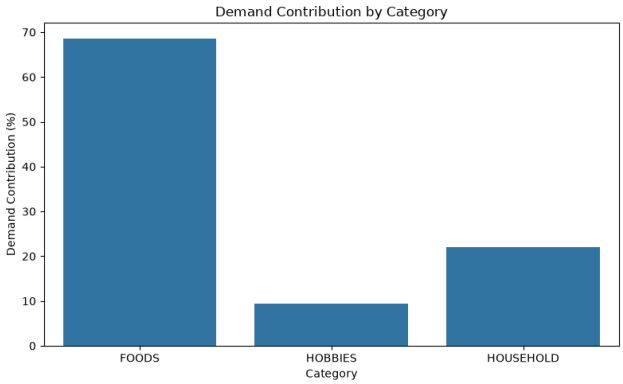


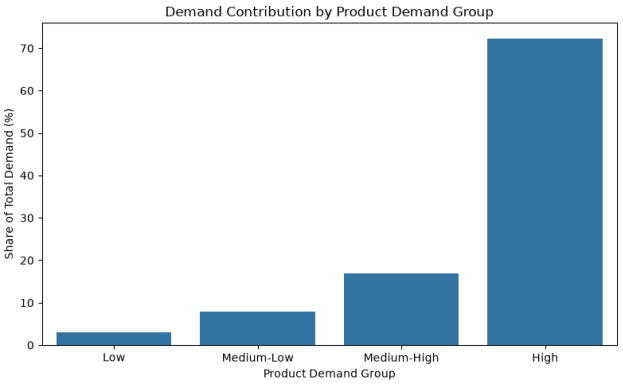
Demand increased over the historical period, but short-term variation remained substantial.

• Total historical demand: 65.7M units; average daily demand: 34,342 units.

• Average weekend demand was 41,338 units versus 31,533 on weekdays — approximately 31.1% higher.

• The 7-day rolling trend shows an overall upward movement with recurring short-term fluctuations.





• FOODS contributed 68.6% of historical demand, followed by HOUSEHOLD at 22.0% and HOBBIES at 9.3%.

• The highest-demand 25% of products contributed approximately 72.3% of total historical demand, showing strong demand concentration.

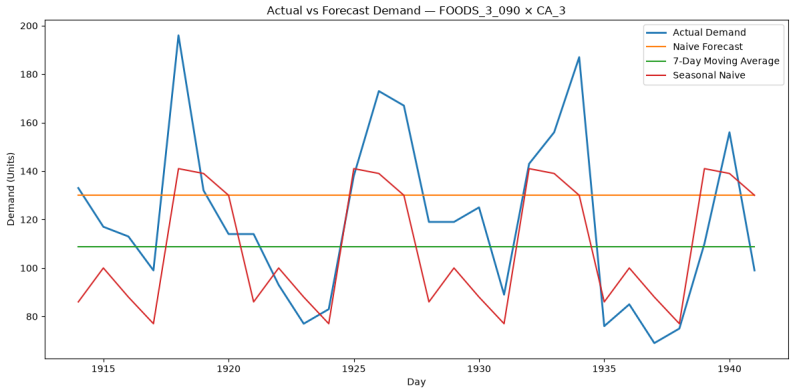
Planning implication: demand is not uniform across time or products, so forecast performance should be examined at relevant segments rather than only at portfolio level.

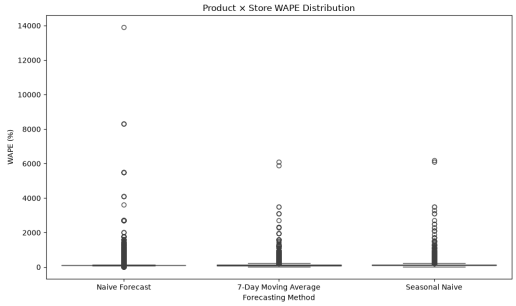
3. FORECAST PERFORMANCE

Three practical baselines were evaluated on the same 28-day Product × Store evaluation period.

Method	WAPE	MAE	Bias
Naive Forecast	95.16%	1.37	+13.19%
7-Day Moving Average	74.04%	1.07	-7.36%
Seasonal Naive	86.22%	1.24	-7.36%

In this 28-day evaluation, the 7-Day Moving Average produced the lowest observed WAPE and MAE among the three practical baselines tested.





• Forecast error distributions were strongly right-skewed across methods, with a substantial number of high-error observations.

• The moving-average baseline still showed materially different performance across Product × Store series.

• This made overall WAPE useful as a portfolio summary, but insufficient for diagnosing where planning problems occurred.

4. WHERE? — Forecast Error Concentration

Forecast performance varies across the Product × Store portfolio.

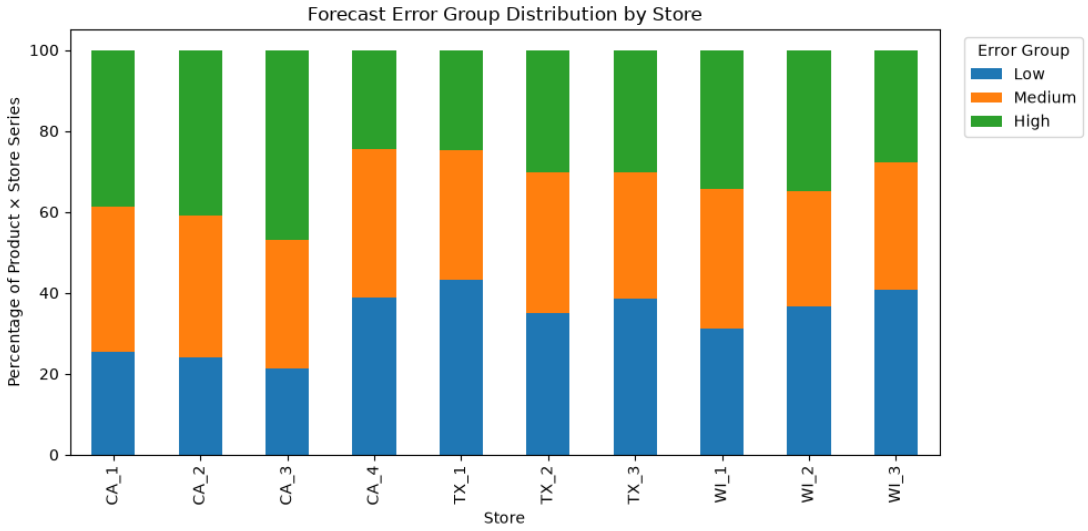
LOW ERROR	*MEDIUM ERROR*	*HIGH ERROR*
10,204 Product × Store series	10,139 Product × Store series	10,147 Product × Store series

High-error series were not evenly distributed across business segments.

• FOODS had the highest proportion of High-Error Product × Store series at 47.43%.

• FOODS_3 had the highest department-level proportion at 50.29%.

• CA_3 had the highest store-level proportion at 47.03%; CA had the highest state-level proportion at 37.73%.

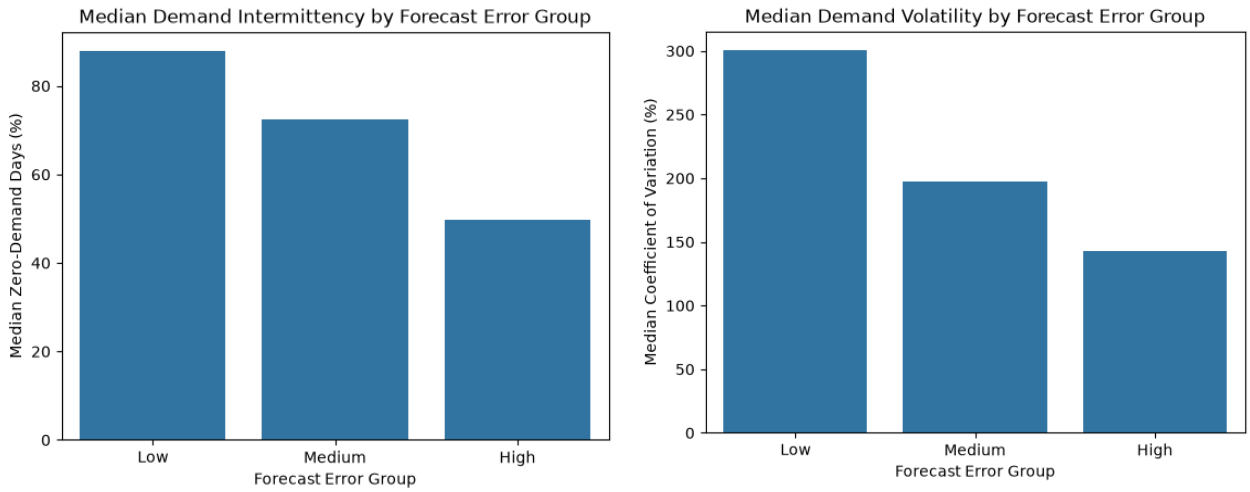


A focused drilldown showed that even within FOODS_3, High-Error proportions varied by store: CA_3 64.52%, CA_1 61.60%, CA_2 58.44%, compared with TX_1 at 36.33%.

Planning implication: portfolio-level forecast metrics can hide concentrated problems; monitoring should identify Product × Store segments with persistent errors.

5. WHY? — What Is Associated With Forecast Error?

The analysis separates absolute error (MAE) from relative error (WAPE), because they describe different planning problems.



• Higher-volume Product × Store series showed higher absolute error (MAE), while their relative error (WAPE) was lower.

• Higher intermittency and higher volatility were associated with higher WAPE but lower MAE.

• Therefore, no single demand characteristic explains forecast error across all error measures.

Demand condition	Observed relationship
Weekday	WAPE was generally higher on weekdays, particularly Tuesday–Thursday, than on weekends.
Within month	WAPE increased from the beginning toward the end of the month.
Event period	Event days had lower WAPE than non-event days during the evaluation period, while event demand was more variable historically.
SNAP period	SNAP-active days had lower WAPE than non-SNAP days across CA, TX and WI during the evaluation period.
Selling price	Higher price groups were associated with higher WAPE but lower MAE.

These are associations observed within the specific 28-day evaluation period; they do not establish causation.

6. BUSINESS DIAGNOSIS

Turning the analysis into a planning story.

HIGHER FORECAST ERROR

Product × Store performance varies substantially across the demand portfolio.

↓

ASSOCIATED PATTERNS

Intermittency · Volatility · Calendar timing · Demand environment · Segment differences

↓

PLANNING CONSEQUENCE

Overall forecast metrics alone can hide where forecast review is most needed.

What the evidence says

• Forecast difficulty varies by Product × Store, category, department, store and state.

• MAE and WAPE can tell different stories: high-volume series can create larger unit errors without having the highest relative error.

• Calendar and demand-environment patterns are associated with differences in forecast performance.

• The 7-Day Moving Average provided a practical baseline, but performance was not uniform across demand series.

The key analytical skill demonstrated here is not simply producing a forecast. It is using forecast errors to identify **where planning attention should be focused** and then examining the demand characteristics associated with those errors.

7. WHAT TO DO? — Planning Implications

Recommendations are limited to actions supported by the analysis.

01	**Focus review on persistent Product × Store errors** Regularly review products and stores where forecasts repeatedly differ significantly from actual sales.
02	**Give more attention to hard-to-predict products** Prioritize products with sparse, irregular or highly variable demand for closer forecast review.
03	**Use both units and percentage differences** Consider MAE and WAPE together so high-volume and low-volume products are not interpreted the same way.
04	**Consider recurring calendar patterns** Pay closer attention to forecast performance around weekdays and within-month periods where larger differences were observed.
05	**Use the recent 7-day average as a practical baseline** Use the 7-day average as a starting point for short-term planning and monitor where it performs poorly.
06	**Investigate persistent forecast problems** When errors persist, investigate promotions, product availability, assortment changes and other business events before changing the forecasting approach.

Portfolio takeaway

The project demonstrates an end-to-end analytics workflow: validate and transform large retail data, identify demand patterns, build practical forecasts, measure forecast performance, diagnose segment-level error, and translate evidence into focused planning actions.